



ELECTROWITS



***Utilization of
Science &
Mathematics.
Applied to
Practical
Problems in the
Field of
Communications.***

INSIDE THIS ISSUE:

Inauguration	2
Value Added Courses	4
Academic Achievements	6
Industrial Visits	9
Student Corner	10
Faculty Corner	14
Clippings	6

Electronics and Communication Engineering

The Department of Electronics and Communication Engineering (ECE) is established in the year 2013 with a vision to develop competent professionals to meet the global challenges in modern engineering society and to educate the students to become enlightened engineers with latest know-how to take on the challenges concerned with the society. At present RIT offers a 4-year B.E. ECE programme with a sanctioned intake of 120 students. The Laboratories are equipped with latest technology in the fields of Electronic Devices and Circuits, Digital Electronics, Linear Integrated Circuits, Microprocessors & Microcontrollers, Very Large Scale Integration Circuits and Digital Signal Processing that offer opportunities on a wide range of hardware and advanced software packages. The Department encompasses professional associations such as Students Technical Association "Electronica", IETE, ISTE chapters which provide students the platform to learn professional and soft skills.

Our Department is stepping into the third year (2015-2016) with the great vision and goals. In the academic year 2014-2015, the Department had organized various Value Added Courses, Conference, Faculty Development Training Programme, Guest lectures and Industrial Visits. The students had excelled in both co-curricular and extra curricular activities. The students had participated in various competitions and won prizes. They produced a pass percentage of 96% in the first semester university examination 2014. The Department wishes to release the news bulletin "ELECTRO WITS" for every period of time from now onwards.

EDITORIAL BOARD

Editor - In - Chief

Dr.S.Periyanayagi HOD I/C

Editor

Ms.S.S.Keerthika Asst. Prof

Student Editors

**Ms.R.Veeralakshmi, II Year ECE
Ms.S.J.Poornima, II Year ECE**

ELECTRONICA Inauguration



INAUGURAL ADDRESS BY THE CHIEF GUEST

Our Association " ELECTRONICA 2014 " was inaugurated on 23.07.2014 in the ECE Seminar Hall by Dr.K.Venkatraj, Principal, PACR Polytechnic College, Rajapalayam.

Mrs.B.Deepa Lakshmi, AP (SG)/ Head Incharge welcomed the gathering. The annual plan of the association activities was presented. The chief guest motivated the students to create interest to improve their technical knowledge by experimenting the latest technologies like Arduino Boards, Portable 8051 microcontrollers and multipurpose robotic boards etc. apart from their curriculum.

TIC

*"Pick your Ideas
and
Click It"*

Technical Incubation Centre

The Technical Incubation Centre was inaugurated by our principal, Dr. R.V.Mahendra Gowda on 27th August 2014 . It is equipped with various platforms such as printed circuit board design, arduino boards and various software environments. The intension of the TIC is to enhance the practical knowledge of the students in various fields of Electronics and Communication Engineering.

Fresher's Day Programme

The RIT organized a Fresher's Day Programme on 05.08.2014. The Department organized Students Interaction Programme on 06.08.2014, which made a platform for the new batch of ECE students to come out of the shell and to interact .

Blood Donation Camp

Organizing blood donation camps is the perfect way to cater to the demand of blood. "Everyone wants to contribute towards the society and save lives", Keeping this as a our motto. Our RIT NSS team organized blood donation camp on 23.08.2014 in order to bring awareness among the students regarding social responsibilities. From the department, II year NSS volunteer students eagerly participated and donated blood.



Faculty Development Programme on Introspection of Teaching Learning Process

The Two Days Faculty Development Programme entitled on the theme "Introspection of Teaching Learning Process" was organized on 23.12.2014 & 30.12.2014. It was a platform for improving the teaching and learning process for all intra college faculties. The various methodology of teaching, art of writing, presentation, pattern of question paper setting, presentation skills, role of class adviser, use of modern teaching aids such as interactive projector and google classroom were delivered by the internal faculties of RIT.

- * Dr.S.Periyanayagi handled the topic "Leading by Examples"
- * Mrs.B.Deepa Lakshmi handled the topic "Effective Handling of Practical Sessions"

***Learning
is a
Continuous
Process***

Nature Club

In order to protect our environment, RIT has initiated and inaugurated a Nature Club on 06.08.2014, devoted for spreading awareness and shedding light on the various environmental issues which are resulting in major catastrophes all over the world creating chaos and destruction. To curb the practices done against the environment, the nature club has taken great strides of awareness in reaching out to the students which emphasized on the urgent need in taking measures to protect our mother nature and make our way of living as "eco-friendly" as possible. The students of I year and II year are active members of this club.

Guest Lecture

To sharpen the minds of young people, Mr.A.K.Mohammed Azad, CEO, REACH Information & Communication Systems, Coimbatore has delivered a guest lecture on 26.03.2015. He inspired the students with his great talk on the topic "Welcome to the World of Electronics and Communication Engineering". about the recent issues in communication field with lots of practical solutions.



***"Expect
Transforms
their Ideas to a
Budding
Experts"***

Value Added Course on ARDUINO



In order to bridge the gap between theoretical and practical knowledge, the Department had organized five days value added course on "Arduino" for II year ECE students from 04.12.2014 to 08.12.2014. The program started with an inaugural function, then lecture sessions handled by the resource persons Mr.G.Jai Sarathy and Mr.S.Karthikeyan from One Yes Technologies, Chennai. They gave detailed explanation about every component related to Arduino Board and the installation procedure of Proteus 7 IDE software and made the students to draw simple circuits and run the simulation using Proteus.

Students had an exposure with the hands on training about uploading different programs to the arduino board and testing the output for blinking LED, Buzzer and motor driver, robotics like line follower, Obstacle detection using two types of sensors and advanced motors using arduino board.

*"It
Create the
Bonding
Between the
Paper
and the
Devices..!"*



ARDUINO COURSE

Five Days Value Added Course on ARDUINO

Value Added Course on MATLAB

In order to enrich the software skills of the students, the Department had organized a five days value added course on "MATLAB Software" from 26.05.2015 to 30.05.2015, for II year ECE students. The program started with an inaugural function. Dr.S.Periyanayagi, HOD/Incharge, motivated the students to improve their practical skills by attending these types of value added courses in her inaugural address. Then the lecture sessions were handled by Mr.Moorthy Kanagaraj and Mr.M.Kannan, Product Engineer, Enthu Technology Solutions India Pvt. Ltd., Coimbatore.

The trainers guided the students in the hands on training sessions on various areas such as GUI programming, the Raspberry Pi hardware and students implement simple programs in Raspberry Pi hardware using MATLAB software.



MATLAB COURSE

Five Days Value Added Course on MATLAB



Academic Achievements

The academic excellence of our first and second batch students were exhibited in the anna university examinations. The overall pass percentage of the first year A section, first year B section and second year students were 89.55%, 94% & 98.18% respectively.

II Year Toppers

- * M.Divya
- * S.Mareeswari
- * V.Reshma
- * S.J.Poornima
- * G.Aiswarya

I Year Toppers

- * A.Jeya Lakshmi
- * J.B.Anu Selvamathi
- * M.Harini Jayalakshmi
- * B.Veeramani Viswanathan
- * T.S.Rashmica

Overall Trophy



Mr.N.Lakshmi Narayanan, Vice-Chairman, Cognizant Technology Solutions, Shri.P.R. Ramasubrahmaneya Rajha, Chairman, Ramco Group, Shri.P.R. Venketrama Raja, Vice-Chairman, RIT and Dr.R.V.Mahendra Gowda, Principal, RIT presented the overall trophy for the best academic performance in the College Day Celebration.

National Science Day - Project Exhibition

The national science day was celebrated by organizing the project exhibition on 28.02.2015. The objective of this event is to provide a solid platform for the students to exhibit their hidden talents and creative ideas in the form of models, prototype devices etc. in the science exhibition. It enables the students to acquire practical knowledge through planning, designing, development and hands-on practice. The department had exhibited 26 unique projects which were guided by the following faculties;

- ♦ Dr.S.Periyanayagi
- ♦ Mrs.B.Deepa Lakshmi
- ♦ Mr.K.Ragavan
- ♦ Mr.A.Azhagu Jaisudhan Palani

List of Projects Exhibited

High Speed Photography Trigger	Clap Switch
Electronic Eye Controlled Security System using LDR	Controlling Home Appliances Using Remote
Automatic Washroom Light Switch	Anti—Theft Alarm for Vehicles
Automatic Railway Gate Controller	Rain Alarm
Electronic Toy Piano Using 555 timer	LED Cube 4*4*4
Hybrid Pump	Stress Meter
Automatic On- Off Night Lamp	Formation of Waves Using Pendulum
Anti—Theft Alarm for Two Wheelers Using Touch Sensor	Car Parking Guard Circuit Using Infrared Sensor
Door Opening Alert Circuit	Pull Pin Security Alarm
Auto Irrigation System	Metal Detector
Speed Controller Of Fan	Automatic Street Light
Doorphone Intercom	Water Level Indicator
Mobile Charger	Electronic Letter Box





Implant Training

In order to cope with the rapidity with which knowledge itself changes, today's students must seek to achieve theoretical and practical academic excellence. To bridge the gap between theoretical and practical knowledge in our technical areas, our students had undergone implant training in various organizations in the field of electronics and communications during their semester holidays. This will provide an industrial exposure to the students as well to develop their career in the high tech industrial requirements. The students had undergone training in

- * Bharat Sanchar Nigam Limited
(Virudhunagar, Tirunelveli, Tuticorn, Madurai & Chennai)
- * Railnet Software Solutions

National Conference on Innovations in Engineering, Science and Technology (NCIEST)

The RIT has organized the first National Level Conference on 6th & 7th March 2015 with the aim of creating a unique opportunity for engineering students, faculty members and researchers from various disciplines to share their expertise in a broad spectrum of innovative topics in emerging technologies. This conference opens the platform for both internal and external participants. It provided an opportunity to interact and share their knowledge and ideas with their peers, to create new technical associations and partnerships in areas pertaining to the fields of engineering and technology. The II year students had presented 16 paper from the various areas in the discipline of Electronics and Communication Engineering.

Industrial Visit for II year



Vikram Sarabhai Space Centre, Thiruvananthapuram

67 Students of II Year ECE visited VSSC on 06.02.2015. They had a glimpse of the development of satellite launch vehicles and distinct technology domains to achieve a high tech realm of vehicles technology.

Industrial Visits for I Year



Sri Ramco Spinners, Rajapalayam

105 Students of I Year ECE visited "Sri Ramco Spinners", on 07.02.2015. They had exposed to the world's best machinery and sophisticated electronic instruments for high quality combed cotton yarn.

Prize Winners for Co-Curricular Activities

***"Winners
never
quit
and
quitters
never win...!"***

V.S.Iasikha, II Year Electronics and Communication Engineering participated and secured First position in Zonal Round of RoboTryst-2014 National Level Championship at PACR Polytechnic College organized by Robosapiens Technologies Pvt. Ltd. In association with Tryst 2014 IIT, Delhi.

V.Reshma, II Year Electronics and Communication Engineering participated and sort listed for delivering a talk in the Preliminary round of ICTACT Youth Talk 2014, Organized by in ICT Academy of Tamil Nadu.

M.Saranyamalini, S.J.Poornima, of II Year Electronics and Communication Engineering has participated and secured second prize in the events of Master Minds in "Gyan Mitra'15", a National Level Techno-Management Symposium organized by Mepco Schlenk Engineering College, Sivakasi on 21st February, 2015.

K.Abirami, II Year Electronics and Communication Engineering has secured First prize in Inter Collegiate Creative Writing 2015 organised by Blue Sky of Mepco Schlenk Engineering College, Sivakasi on 21st February, 2015.

M.Menaka, T.Kaviya of II Year Electronics and Communication Engineering has participated and secured Third prize in the event HIT THE HINT and POWER CLASH conducted during Orcle'15 a National level Technical Symposium in Government College of Technology, Coimbatore on 19th & 20th March, 2015.

S.Aishwarya, V.S.Abinaya Shree of II Year Electronics and Communication Engineering has Secured First Prize in the event of "Technical Quiz" as a part of E'Seigniors'15- National Level Technical Symposium organized by SACS MAVMM Engineering College, Madurai on 20th March, 2015.

S.Aishwarya, V.S.Abinaya Shree of II Year Electronics and Communication Engineering has presented a Paper and Secured Third Prize in the event as a part of E'Seigniors'15- National Level Technical Symposium organized by SACS MAVMM Engineering College, Madurai on 20th March, 2015.

Student's Participation

K.Ramalakshmi, M.Saranyamalini, M.Parvathi, S.J.Poornima, C.Mathuri Banu, S.V.Iasikha, V.Reshma, of II Year Electronics and Communication Engineering has attended the Workshop on "Intellectual Property Rights (IPR) Awareness Programme", Organized by Department of Mechanical Engineering, Ramco Institute of Technology on 23rd August, 2014.

M.Saranyamalini, S.J.Poornima, S.V.Iasikha, M.Divya, of II Year Electronics and Communication Engineering has Participated in the events of "Gyan Mitra'15", a National Level Techno-Management Symposium organized by Mepco Schlenk Engineering College, Sivakasi on 21st February, 2015

M.Harini Jayalakshmi, I Year Electronics and Communication Engineering has participated in the events of "Gyan Mitra'15", a National Level Techno-Management Symposium organized by Mepco Schlenk Engineering College, Sivakasi on 21st February, 2015.

B.Veeramani Viswanathan, P.Premkumar of I Year Electronics and Communication Engineering has Participated in Quiz Competition Conducted by G.Venkataswamy Naidu College, Kovilpatti on 4th February, 2015

V.S.Iasikha, II Year Electronics and Communication Engineering has participated in the Final rounds of RoboTryst-2015, A National Level Championship Organized by Robosapiens Technologies Pvt. Ltd. In association with Tryst 2015 at IIT, Delhi held on 27th Feb-2nd March 2015 during Tryst-2015 Annual Science, Technology and Management Festival of IIT, Delhi.

A Two day National Conference on Innovations in Engineering, Sciences & Technology (NCIEST-2015) was organized at Ramco Institute of Technology on 06th & 07th March, 2015 Top 10 students of I & II Year Electronics and Communication Engineering students participated.

N.Rajalakshmi, of II year Electronics and Communication Engineering has presented a project entitled "Hybrid Pump" in State Level Mini Project Contest-M/C 2015 conducted on 25th March 2015 organized by the Department of Mechanical Engineering.

***"Nothing ever
becomes
real
till it is
experienced...!"***

SPORT'S ACHIEVEMENTS

Our students had contributed equally in both academic and extra curricular activities. The I and II year students involved in various sports events and won many prizes organized by the Department of Physical Education.

***"Winners
don't do
different
things
but they do
things
differently."***

S.No.	Name of the Students with Year	Name of the Event	Prizes Won
1	V.Reshma, S.Deepika / II Year and M.Harini	400mts Relay	First
2	K.Nivethapriya and B.Muthu Divya Gayathri /	400mts Relay	Second
3	S.J.Poornima, C.Sathyasree, M.Saranyamalini /	400mts Relay	Third
4	C.Sathyasree / II Year	Long Jump	Second
5	K.Nivethapriya / I Year	Shot put	Second
6	V.Reshma / II Year	Javelin Throw	Third
7	S.J.Poornima / II Year	Hammer Throw	Third
8	R.Stefy / I Year	Carrom	First
9	M.Dineshkumar / I Year	400mts	Second
10	M.Divya / II Year	400mts	Third
11	S.Deepika / II Year	800mts	First
12	Nivetha Priya / I Year	100mts	First
13	Harini Jayalakshmi and Nivetha Priya / II Year	200mts	First

PAPER PRESENTATIONS

1. K.Ramalakshmi & M.Swarnalaxhmi of II Year has Presented a Paper titled "Office Automation and Security System using Android Technology" in the National Level Technical Symposium, ENIAC'15 on 13th March 2015 organized by Kalasalingam University, Krishnankovil.

2. S.Aishwarya & V.S.Abinaya Shree of II Year has presented a Paper titled "Digital Jewellery " and Secured Third Prize in the event as a part of E'Seigniors'15- National Level Technical Symposium organized by SACS MAVMM Engineering College, Madurai on 20th March, 2015.

3. K.Ramalakshmi & M.Swarnalaxhmi of II Year has Presented a Paper titled "Office Automation and Security System using Android Technology" in Two days National Conference an Innovations in Engineering, Sciences & Technology (NCIEST-2015) organized by Ramco Institute of Technology, Rajapalayam on 06th & 07th March, 2015.

4. M.Divya & A.Beena of II Year has Presented a Paper titled "Human Computer Interaction System Based on Hand Gestures" in Two days National Conference an Innovations in Engineering, Sciences & Technology (NCIEST-2015) organized by Ramco Institute of Technology, Rajapalayam on 06th & 07th March, 2015.

5. V.Reshma & M.Parvathi of II Year has Presented a Paper titled "Deep Brain Stimulation (A Power-Efficient Wireless System with Adaptive Supply Control)" in Two days National Conference an Innovations in Engineering, Sciences & Technology (NCIEST-2015) organized by Ramco Institute of Technology, Rajapalayam on 06th & 07th March, 2015.

***" Under
adverse
conditions -
some people
break down,
some
break records .! "***

Faculty Corner

PH.D AWARDED

Dr.S.Periyanayagi has been awarded Ph.D on the thesis entitled "SWARM BASED DEFENCE TECHNIQUES FOR CERTAIN ATTACKS IN IEEE 802.15.4 BASED WIRELESS SENSOR NETWORKS" by Anna University, Chennai in the year of 2014.

Publications in Conferences

S.Periyanayagi presented a paper on "A Defense Techniques Based on SI Model for Certain Attacks in IEEE802.15.4 Based WSN", Two day National Conference an Innovations in Engineering, Sciences & Technology (NCIEST-2015) organized by Ramco Institute of Technology, Rajapalayam on 06th & 07th March, 2015.

K.Ragavan, G.Thiruselvi, M.Ramaiah presented a paper on "Comparative Analysis of SVM and BPA for Wheelchair Detection", Two day National Conference an Innovations in Engineering, Sciences & Technology (NCIEST-2015) organized by Ramco Institute of Technology, Rajapalayam on 06th & 07th March, 2015.

K.Ragavan, M.Ramaiah presented a paper on "Design Hilbert-Pair of Wavelets", Two day National Conference an Innovations in Engineering, Sciences & Technology (NCIEST-2015) organized by Ramco Institute of Technology, Rajapalayam on 06th & 07th March, 2015.

A.Azhagu Jaisudhan Pazhani, C.Vasanthanayaki presented a paper on "Design of Multi output binary adder using modified parallel prefix Addition" in the Two day International Conference on "Electrical, Electronics, Instrumentation and Computer Communication organized by Karpagam College of Engineering, Coimbatore during 13th December, 2014.

*" The noblest
search
is the
search
for
excellence...! "*

Publication in Journals

1. A.Azhagu Jaisudhan Pazhani, C.Vasanthanayaki published a paper titled "Design of Multi output binary adder using modified parallel prefix Addition", in the International journal of Advances in Natural and Applied Sciences, 2014.

2. A.Azhagu Jaisudhan Pazhani, C.Vasanthanayaki published a paper titled "FPGA Implementation of computationally Efficient Modified Parallel Prefix Adder" in the International Journal of Applied Engineering Research, ISSN 0973-4562 Vol.10 No.1(2015) pp.367-371.

Events Attended

- * Mrs.G.Thiruselvi participated in five days Faculty Updating Programme on "PCB Design Techniques using CAD Packages" during 27th July 2014 to 10th August 2014 was conducted by National Institute of Electronics and Information Technology, New Delhi at P.A.C.R Polytechnic College, Rajapalayam.
- * Mrs.B.Deepa Lakshmi has participated in a two week ISTE workshop on "Control Systems" conducted by Indian Institute of technology, Kharagpur from 2nd to 12th December, 2014 held at PSG College of Technology, Coimbatore.
- * Mr.K.Ragavan has participated in UKIERI Sponsored two day workshop on Advances in Cognitive Radio and its Hardware Implementation organized by the Department of ECE, NIT, Trichy during 23rd & 24th December, 2014.
- * Mr.K.Ragavan has participated in the seven days FDP on "Computer Networks" at RIT, Rajapalayam from 15th to 21st December 2014.

Valedictory Function

The ELECTRONICA association activities for this academic year come to the end with the valedictory function on 26.03.2015. The chief guest for this function is Mr.A.K.Mohammed Azad, CEO, REACH Information & Communication Systems, Coimbatore. He inspired the students with his great talk about the recent issues in communication field with lots of practical solutions.

***" This ending
will have
a
new
beginning .! "***

News Paper Clippings

- ★ கருத்தரங்கு, , ராம்கோ இன்ஸ்டிடியூட் ஆப் டெக்னாலஜி, ராஜபாளையம், காலை 9 மணி.
- ★ சிறப்பு பயிற்சி, ராம்கோ இன்ஸ்டிடியூட் ஆப் டெக்னாலஜி, ராஜபாளையம், காலை 9.30 மணி

இந்த நேரத்தில் கருத்தரங்கு ராம்கோ இன்ஸ்டிடியூட் ஆப் டெக்னாலஜி, ராஜபாளையம், காலை 9 மணி பங்கேற்பு, டெக். சேல்ஸ் துணைபுரத்து மேலாளர் டி. ஆர். ராமசுவாமி, ஏற்பாடு, கோயம்புத்தூர் கோட் டெக்னாலஜி பள்ளி.
 அரங்கேற்றம் கோலா பயிற்சி, ராம்கோ இன்ஸ்டிடியூட் ஆப் டெக்னாலஜி, ராஜபாளையம், காலை 9.30 மணி, தலைமை: ராம்கோ இன்ஸ்டிடியூட் ஆப் டெக்னாலஜி முதல்வர் மகேந்திர சுந்தர், ஏற்பாடு, டெக்னாலஜி வன் டெக்னாலஜி

• RAJAPALAYAM

Ramco Institute of Technology:
Valedictory function of Students
Technical Association
'ELECTRONICA', chief guest A.K.
Mohammed Azad, CEO, REACH
Information and Communication
Systems, Coimbatore, ECE
Seminar hall, 10 a.m.

• RAJAPALAYAM

Ramco Institute of Technology:
Department of Mechanical
Engineering, nine-day international
certification programme on
Autocad for 11 year students
organised by CADD Technologies
School of Design, Coimbatore.
T.R. Ramsankar, DGM-Tech
Sales, resource person, 9 a.m.;
Department of Electronics and
Communications Engineering,
five-day training programme
conducted by One Yes
Technologies, Chennai, for 11 year
students, 9.30 a.m.

For Further Details

www.ritrjpm.ac.in

Published by:
DEPT. OF ELECTRONICS & COMMUNICATION ENGG.
RAMCO INSTITUTE OF TECHNOLOGY
North Venganallur Village
Rajapalayam - 626117
Virudhunagar Dt,
Tamil Nadu

Contact:

Phone: 04563-233400 (99 lines)
Fax: 233499
e-mail : rit.rjpm@gmail.com
web site : www.ritrjpm.ac.in